TP53RK phosphorylates TP53

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Introduction

Reactome is open-source, open access, manually curated and peer-reviewed pathway database. Pathway annotations are authored by expert biologists, in collaboration with Reactome editorial staff and cross-referenced to many bioinformatics databases. A system of evidence tracking ensures that all assertions are backed up by the primary literature. Reactome is used by clinicians, geneticists, genomics researchers, and molecular biologists to interpret the results of high-throughput experimental studies, by bioinformaticians seeking to develop novel algorithms for mining knowledge from genomic studies, and by systems biologists building predictive models of normal and disease variant pathways.

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Literature references

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Reactome database release: 98

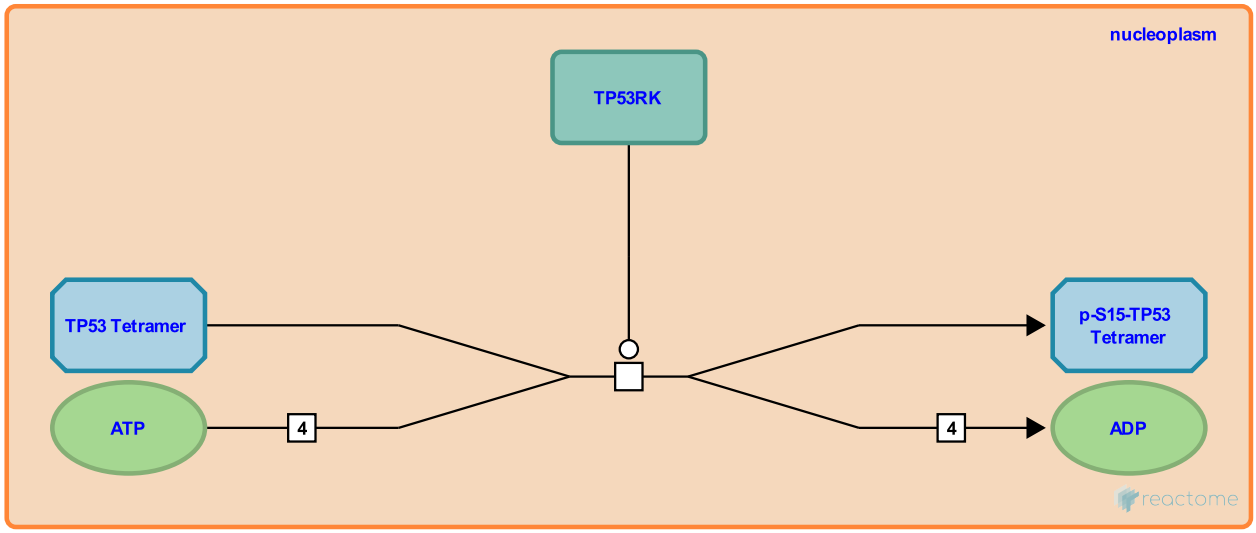
This document contains 1 reaction (see Table of Contents)

TP53RK phosphorylates TP53

Stable identifier: R-HSA-6805479

Type: transition

Compartments: nucleoplasm



The atypical protein serine/threonine kinase TP53RK (TP53-regulating kinase), also known as PRPK (p53-related protein kinase), phosphorylates TP53 (p53) on serine residue S15 (Abe et al. 2001, Facchin et al. 2003).

Literature references

Abe, Y., Matsumoto, S., Wei, S., Nezu, K., Miyoshi, A., Kito, K. et al. (2001). Cloning and characterization of a p53-related protein kinase expressed in interleukin-2-activated cytotoxic T-cells, epithelial tumor cell lines, and the testes. *J. Biol. Chem., 276*, 44003-11.

Facchin, S., Lopreiato, R., Ruzzene, M., Marin, O., Sartori, G., Götz, C. et al. (2003). Functional homology between yeast piD261/Bud32 and human PRPK: both phosphorylate p53 and PRPK partially complements piD261/Bud32 deficiency. *FEBS Lett., 549*, 63-6.

Editions

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